

Answer: D

**10** Heat supplied to system by heater = heat gained by water + heat gained by tank

$$P \times t = mc\Delta\theta + ml_v + C\Delta\theta$$

$$5.2 \times 10^3 \times t = [14 \times 4200 \times (100 - 30)] + (14 \times 2.26 \times 10^6) + [6500 \times (100 - 30)]$$

$$t \approx 7000 \text{ s}$$

Answer: D

**11**  $pV = nRT$